ERP System Design for Manufacturing

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Section - 2

Industry Overview
This assignment examines an ERP system for a manufacturing company, with industrial machinery and components. They operate multiple plants with supply chains, multi-stage assembly, and inventory challenges. Bills of materials run deep, work orders are constantly flowing, and everything comes with regulatory overhead. A mid-sized to large manufacturer typical of the sector.

Required ERP modules

Five modules matter most. Here's the ranking:

Rank	Module	Why It Matters
1	Manufacturing	Handles production planning, bills of materials, work orders, shop-floor scheduling. Without this, nothing gets made.
2	Inventory Management	Tracks raw materials, work-in-progress, finished goods across warehouses. No visibility here means no accurate costing.
3	Procurement	Manages suppliers and purchase orders. Ties directly to production schedules and material availability.
4	Finance & Accounting	Records production costs, GL entries, payables, receivables. Needed for accurate product costing and financial statements.
5	Sales	Processes customer orders and quotations. Demand signals flow straight into production planning.

Key master data

Before anything can transact, certain data has to exist in the system:

Master Data	What It Is	Why You Need It
Bill of Materials (BOM)	The recipe for each product, all components and sub-assemblies with exact quantities.	Without a BOM, you can't tell the system what to build.
Item Master	Catalog of everything: raw materials, components, sub-assemblies, finished goods. Includes unit of measure and warehouse location.	Foundation for tracking inventory and fulfilling orders.
Work Centers & Resources	Definition of production equipment, labor pools, capacity (machines, lines, quality stations).	Needed to schedule production and allocate operation costs.
Supplier Master	Vendor info: payment terms, lead times, quality ratings, supply agreements.	Determines which supplier to order from and how long to wait.
Routing / Process	Exact steps to manufacture each product. Which work center, how long, what resources.	Tells the shop floor what to do and how to cost each operation.
Cost Centers & GL Accounts	Organizational structure and accounting codes for production and overhead.	Needed to track costs and produce financial reports.
Warehouse & Location	Physical storage areas (raw materials, WIP staging, finished goods dock).	Required for accurate inventory visibility.
Customer Master	Customer account details, delivery addresses, credit limits, pricing agreements.	Foundation for processing sales orders and customer-specific rules.

Core transactions

Three transactions happen constantly and drive everything:

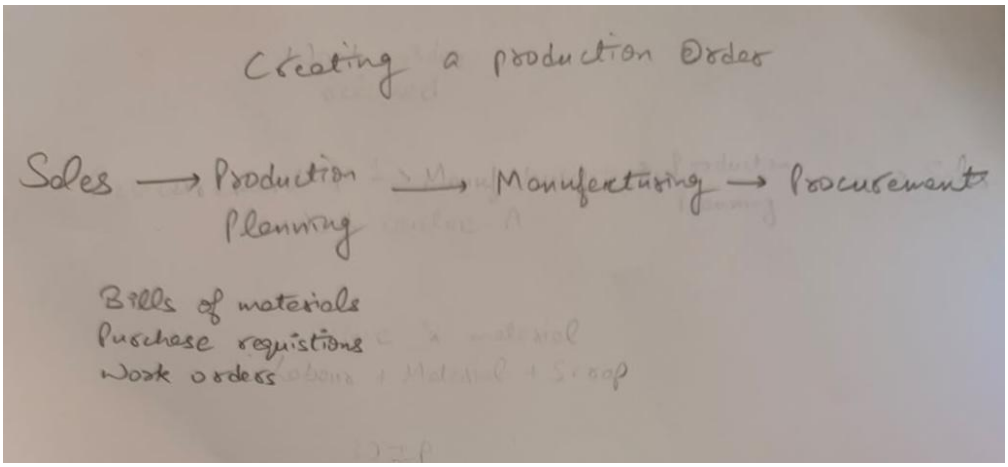
1. Creating a production order

A sales order or forecast comes in. The planning department creates a production order that immediately:

Explodes the bill of materials to show every component needed

Creates purchase requisitions for external suppliers

Schedules work orders across production resources



2. Receiving materials

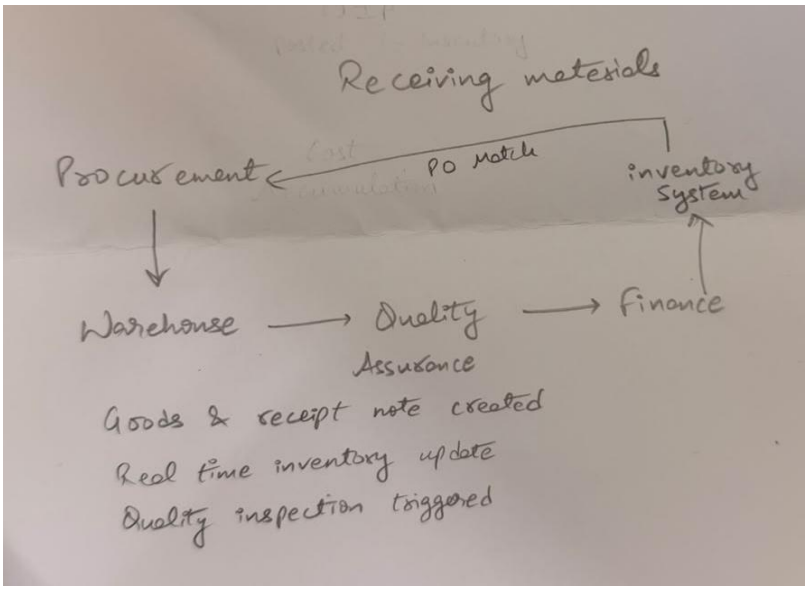
When goods arrive from a supplier, the warehouse records them against the PO. This transaction:

Records a goods receipt note (GRN)

Updates inventory in real-time

Triggers quality inspection

Posts to the general ledger (inventory asset and accounts payable)



3. Recording production progress

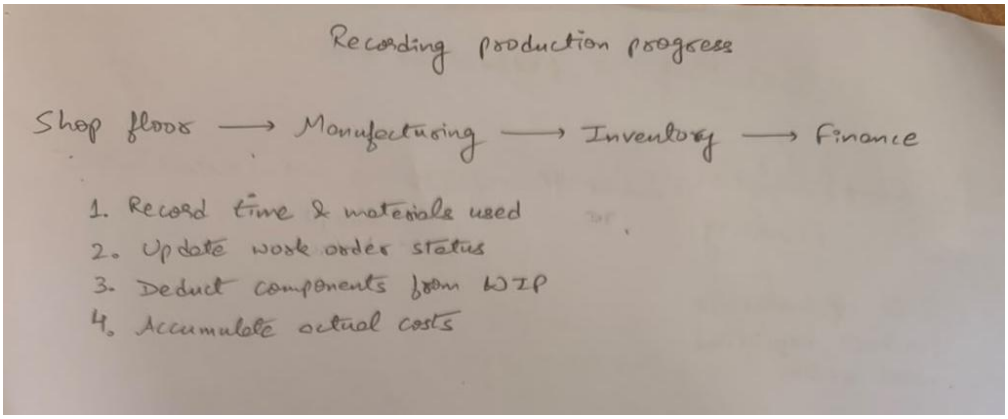
As work moves through each production step, the shop logs what was done. This transaction:

Records time and materials used at each work center

Updates work-in-progress inventory

Triggers the next step

Accumulates actual costs for later analysis



Critical workflow: Production order approval

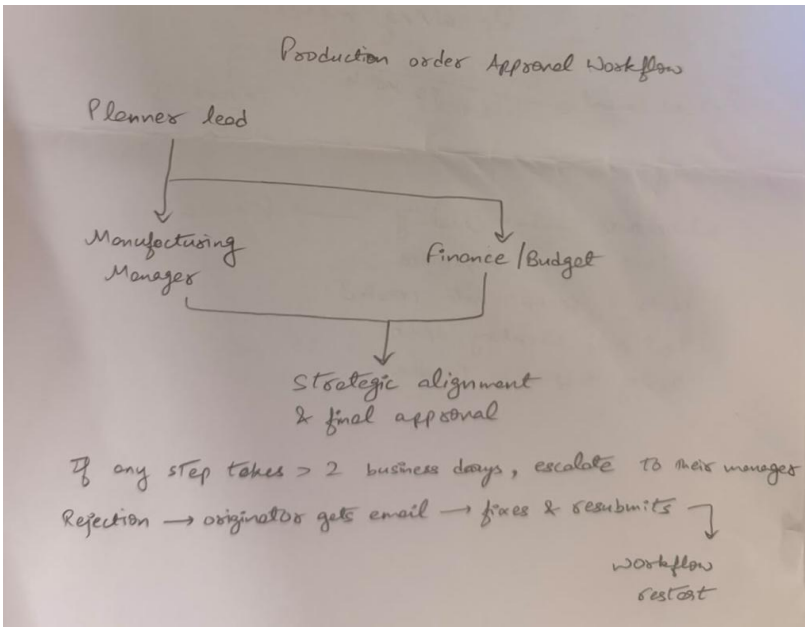
Large production orders need approval before materials are purchased. This catches problems early and keeps costs in check.

The approval process

Step	Approver	Checking For	If Rejected
1	Production Planner Lead	Can we actually make this? Materials and work centers available? Timelines realistic?	Return to originator for fixes
2	Manufacturing Manager	Do we have shop-floor capacity right now? Can this fit into the schedule?	Return to planner with constraints
3	Finance / Budget Controller	Is the estimated cost within budget? Does the margin make sense?	Return to finance with feedback
4	Plant Director	Is this strategically aligned? Is the supplier available? How important is this customer?	Return to finance with objections

How it actually works

Steps 2 and 3 run in parallel once Step 1 approves. If anyone takes longer than 2 business days, their manager gets a notification. The plant director can override the finance step if they document the reason. When an order gets rejected, the submitter gets an email explaining what needs to change. When they fix it and resubmit, the workflow starts over.



Integration points

Two pairs of modules have to work together seamlessly. If they don't, everything breaks.

Manufacturing & inventory must stay in sync

What they exchange:

When production completes a step, manufacturing tells inventory to move materials. Finished goods get received. Components get deducted from raw materials. Scrap gets written off. Every movement carries a lot number for traceability.

What breaks if they don't:

Inventory becomes unreliable. Physical stock never matches the system.

Production stalls. The shop floor can't see what materials are actually available.

Costing fails. You can't allocate material costs to products, so you never know which ones are profitable.

Quality traceability vanishes. If a batch is defective, you can't trace which products contain it.

Procurement & finance must stay in sync

What they exchange:

Procurement creates purchase orders. Finance reserves the budget. When goods arrive, Procurement records the receipt. Finance matches it against the PO and supplier invoice. When everything lines up, Finance pays.

What breaks if they don't:

Invoices go unmatched. You pay the same invoice twice, or pay for something that never arrived.

Budget becomes invisible. You have no idea how much you've actually committed to spend.

Supplier relationships suffer. Payments are late or wrong. Good suppliers stop working with you.

Financial statements are wrong. Payables are overstated or understated. Auditors find problems. Management makes decisions based on bad numbers.

Conclusion

A working manufacturing ERP starts with five core modules, Manufacturing, Inventory, Procurement, Finance, and Sales. But the real work is in the details: master data accuracy, workflows that catch problems, and modules that actually talk to each other.

Production order approval is one example of how workflow prevents expensive mistakes. The production order lifecycle, from creation through approval to shop-floor execution and costing, shows how the pieces connect. When Manufacturing and Inventory sync, the shop floor knows what to do. When Procurement and Finance sync, the company stays solvent and suppliers keep delivering. The system only works when all the parts work together.